in Lhasa without permission. After this incident, Nain Singh seldom appeared in public. At night, he would climb out quietly from the window onto the roof of the small inn where he stayed. Then, he would use his sextant to determine latitude by measuring the angular altitude of the stars. He also used his thermometer to record the boiling point of water as the higher the altitude, the lower the boiling point. Using this method, he estimated that Lhasa was at an altitude of 3420 metres above sea level. This is very close to the modern measurement of 3540 metres.

Nain Singh left Lhasa in April along with a Ladakhi caravan and headed west for 800 km along the River Tsangpo. All along he kept taking readings in secret. After two months with the caravan he slipped away on his own and made his way back to India via the sacred Mansarovar Lake. He arrived back at the Survey of India headquarters on 27 October 1866. During his twenty-one-month adventure, he had surveyed thousands of kilometres, taken thirty-one latitude fixes, and determined elevation in thirty-three places and the first accurate position of the Tibetan capital. Nain Singh would return to Tibet in 1873–75 to explore a more northerly route from Leh in Ladakh to Lhasa. Some of his family members would join the dangerous profession and work for the Survey of India.

Nain Singh's reports raised an interesting geographical question. Where did the Tsangpo flow? Did it cross the Himalayas as Singh suggested, and become the river known to Indians as the Brahmaputra? In order to solve the mystery, the Survey decided to slip someone back into Tibet and float something identifiable down the Tsangpo. If it turned up in the Brahmaputra in Assam, they would know the answer.

The two man team for the job consisted of a Chinese lama living in Darjeeling and a Sikkimese surveyor called Kinthup. The Survey had badly misjudged the lama, who was more intent on enjoying himself than on getting the work done, and often got drunk. The team was stuck in one village for four months because the lama fell in love with their host's wife. When the affair became known, the lama had to pay Rs 25 in compensation and leave. Things did not improve when at last they had crossed into Tibet. The lama sold Kinthup as a slave to the headman of a Tibetan village and